Lattice surgery in Stim: measuring $Z_A Z_B$ between two surface-code patches

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https://github.com/DerivedUsername/lattice-surgery-stim

1 Problem

The rotated surface code stores one logical qubit in a $d \times d$ grid of data qubits; checks of weight 2 or 4 are measured on the faces, and the code space is a joint eigenspace of the checks. Given two patches A and B, we want to measure $Z_A Z_B$ without measuring Z_A or Z_B. This reveals only the parity, so a state such as $\alpha|00\rangle + \beta|11\rangle$ keeps its superposition; it is the basic two-qubit operation between patches. Lattice surgery (Horsman et al., arXiv:1111.4022, §3 and §7.1) does it by merging the patches for d rounds and splitting them again. We build this as a Stim circuit with circuit-level noise, decode it with PyMatching, and compare with two idle patches.

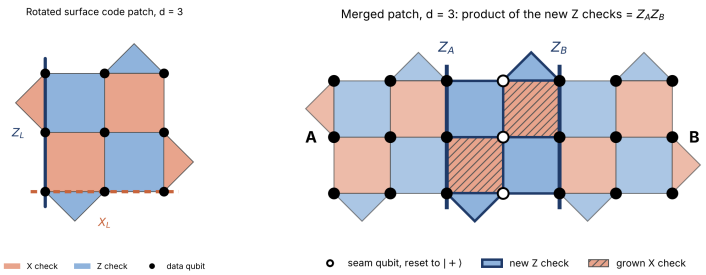


Figure 1: Left: a 3×3 patch. Right: the merged patch, with $d+1$ new Z checks and $d-1$ grown X checks.

2 Why the merge measures $Z_A Z_B$

A column of d seam qubits between the patches is reset to $|+\rangle$, and the checks of the single $(2d+1) \times d$ patch M are measured. Compared with A and B, no check is removed, $d-1$ X checks grow by $X_s X_{s'}$ on two seam qubits, and $d+1$ Z checks are new. An unchanged check returns its old value, and so does a grown one, since $X_s X_{s'} = 1$ on the seam. A new check anticommutes with X_s for a seam qubit s in it, so on every code state its outcome is ± 1 with probability $\frac{1}{2}$ and says nothing about A and B. The product of the new checks is different: each seam qubit lies in exactly two of them, so its Z cancels and

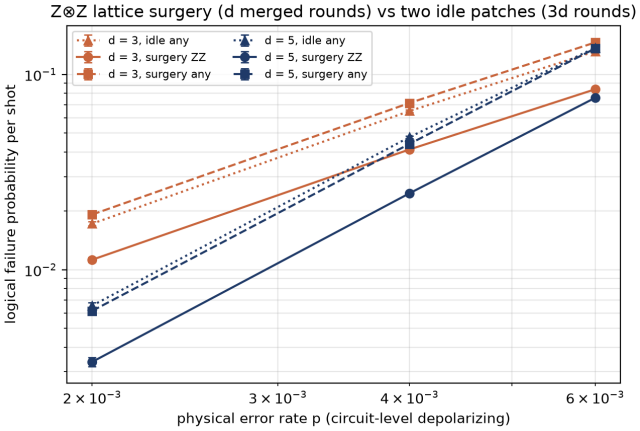
$$\prod_j n_j = Z_{\text{column } d-1} \, Z_{\text{column } d+1} = Z_A Z_B.$$

So the product λ of the $d+1$ random outcomes is the value of $Z_A Z_B$. On the code space the merge acts as $\psi \mapsto U_\lambda P_\lambda \psi$: a projection onto the eigenspace $Z_A Z_B = \lambda$, then an isomorphism onto the code space of M, fixed up to a Pauli correction tracked in software. The operators Z_A and $X_A X_B$ survive as the logical operators of M; X_A and X_B separately do not. Measuring the seam in the X basis splits M back into A and B.

3 Circuit

The package `lattice_surgery_stim` builds the patch geometry, a memory circuit for any set of patches, and the surgery circuit: d rounds on A and B, m rounds on M, d rounds on A and B, and a final Z readout. Detectors compare each check with its previous round; the new checks have none in their first round, and a grown check that shrinks at the split is compared together with its two seam outcomes. Observable 0 is the parity of the new checks in the first merged round, and observables 1, 2 are Z_A, Z_B at readout. The memory circuit matches Stim's `surface_code:rotated_memory_z` in logical error rate at $d = 3, 5$. Stim's shortest undetectable logical error has weight $\min(m, d)$ for $m = 1, \ldots, d$: with fewer than d merged rounds, repeated measurement errors on one new check flip the parity unseen. This is why the merge lasts d rounds.

4 Results



d	p	surg. ZZ	surg. any	idle any
3	0.002	1.12%	1.92%	1.72%
3	0.004	4.13%	7.14%	6.49%
3	0.006	8.39%	14.57%	13.12%
5	0.002	0.34%	0.62%	0.65%
5	0.004	2.46%	4.41%	4.77%
5	0.006	7.59%	13.58%	13.70%

	surgery		idle	
	d=3	d=5	d=3	d=5
qubits	41	109	34	98
rounds	9	15	9	15
detectors	152	744	144	720

Failure rates over 10^5 shots per point, with $m = d$; the baseline is the two patches idle for the same $3d$ rounds. "ZZ" means the parity was mispredicted, "any" that at least one observable was. At $p = 0.002$, $d = 5$ fails about a third as often as $d = 3$; at $p = 0.006$ the two are nearly equal, so the crossing is a little above that. The surgery fails about 10% more often than the baseline at $d = 3$ and is comparable to it at $d = 5$. A possible reason, not tested: during the merge an X chain that flips Z_A must cross the width $2d + 1$ instead of d. The surgery uses $2(2d+1)d-1$ qubits and costs d rounds in time; gates are Z- and X-basis reset and measurement, H and CNOT.

5 Limitations

Only the smooth merge is implemented. Only Z-basis preparation is simulated, so the experiment tests the parity outcome and the survival of Z_A, Z_B, but not the coherence of $X_A X_B$, which needs an X-basis run. Noise is uniform depolarizing with one strength p. Only $d = 3, 5$ and three values of p were run, so no threshold is claimed.